# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 7 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 2

**Full Syllabus (Chapters 1–11)** | RBSE Pattern | Session 2026–27

**Time: 3 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 80**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 20 × 1 = 20

*Choose the correct option and write it in your answer book.*

**1.** For a shadow to form, we need: 1

**(a)** an object only
**(b)** a source of light, an opaque object and a screen
**(c)** a source only
**(d)** a screen only

**2.** The longest day in the northern hemisphere is on about: 1

**(a)** 21 March **(b)** 23 September **(c)** 22 December **(d)** 21 June

**3.** Red litmus turns blue in: 1

**(a)** salt solution **(b)** water **(c)** acid **(d)** base

**4.** When current flows through a wire, the wire becomes: 1

**(a)** hot **(b)** wet **(c)** magnetic only **(d)** cold

**5.** When a metal burns in oxygen, it forms a: 1

**(a)** salt **(b)** non-metal oxide **(c)** acid **(d)** metal oxide

**6.** Rusting needs: 1

**(a)** air and moisture **(b)** only water **(c)** only sunlight **(d)** only air

**7.** Rapid growth of height at puberty is called: 1

**(a)** gestation **(b)** stagnation **(c)** decline **(d)** growth spurt

**8.** Dark-coloured clothes absorb ___ heat than light-coloured clothes. 1

**(a)** more **(b)** equal **(c)** no **(d)** less

**9.** The instrument in a vehicle which records the distance travelled is the: 1

**(a)** thermometer **(b)** odometer **(c)** barometer **(d)** speedometer

**10.** The exchange of gases between the body and the environment is: 1

**(a)** excretion **(b)** digestion **(c)** breathing **(d)** absorption

**11.** Rhizobium bacteria live in the root nodules of: 1

**(a)** grasses **(b)** cereals
**(c)** leguminous plants **(d)** palms

**12.** The bouncing back of light from a surface is: 1

**(a)** absorption **(b)** reflection **(c)** dispersion **(d)** refraction

**13.** The Moon revolves around the Earth in about: 1

**(a)** 27 days **(b)** 365 days **(c)** 1 day **(d)** 12 hours

**14.** Substances that feel soapy and taste bitter are generally: 1

**(a)** acidic **(b)** sour **(c)** basic **(d)** neutral

**15.** Two or more cells joined together form a: 1

**(a)** circuit breaker **(b)** battery **(c)** fuse **(d)** switch

**16.** The property of metals by which they can be beaten into thin sheets is: 1

**(a)** brittleness **(b)** sonority **(c)** malleability **(d)** ductility

**17.** A change in which one or more new substances are formed is a: 1

**(a)** physical change **(b)** change of state

**(c)** reversible change only **(d)** chemical change

**18.** The age group of adolescence is roughly: 1

**(a)** 40 to 50 years **(b)** 11 to 19 years **(c)** 20 to 30 years **(d)** 1 to 5 years

**19.** The three modes of heat transfer are: 1

**(a)** convection, sublimation, condensation **(b)** radiation, freezing, boiling

**(c)** conduction, convection, radiation **(d)** conduction, evaporation, melting

**20.** 1 hour is equal to: 1

**(a)** 360 seconds **(b)** 600 seconds **(c)** 60 seconds **(d)** 3600 seconds

## SECTION — B (Very Short Answer Type) — 6 × 2 = 12

*Answer in about 20–30 words each.*

**21.** Define speed and write its SI unit. 2

**22.** Name the five steps of nutrition in animals. 2

**23.** What is photosynthesis? Write the word equation. 2

**24.** What is the difference between luminous and non-luminous objects? 2

**25.** What are rotation and revolution of the Earth? 2

**26.** Name two common bases used at home. 2

## SECTION — C (Short Answer Type) — 6 × 3 = 18

*Answer in about 40–60 words each.*

**27.** Write the word equation for photosynthesis and explain each term. 3

**28.** Explain with an activity that light travels in straight lines. 3

**29.** Explain how day and night occur on the Earth. 3

**30.** Why should we not taste unknown chemicals to test whether they are acids or bases? 3

**31.** How does a fuse protect an electrical appliance? 3

OR

Why is aluminium used for making foil and cooking vessels? 3

**32.** Describe an activity to show a chemical change using vinegar and baking soda. 3

OR

What is the role of adrenal gland? 3

## SECTION — D (Long Answer Type) — 6 × 5 = 30

*Answer in about 100–150 words each.*

**33.** Describe the activity to show that light travels in a straight line and to produce a shadow, and discuss the factors affecting the size of a shadow. 5

**34.** Explain the rotation and revolution of the Earth with the consequences of each. 5

**35.** Explain acids found in food items and the natural source of each. 5

**36.** What is a battery? Explain how three 1.5 V cells in series produce 4.5 V and what happens to bulb brightness. 5

OR

Describe exceptions in properties of metals and non-metals with examples. 5

**37.** What are the indications of chemical change? Describe three activities. 5

OR

Describe the role of hormones in growth and mention the diseases linked with hormone problems. 5

**38.** Differentiate between conduction, convection and radiation on five points. 5

OR

Describe the oscillatory motion of a pendulum and explain terms: oscillation, time period, amplitude. 5

## END OF QUESTION PAPER — Total: 80 Marks

MASTERMIND